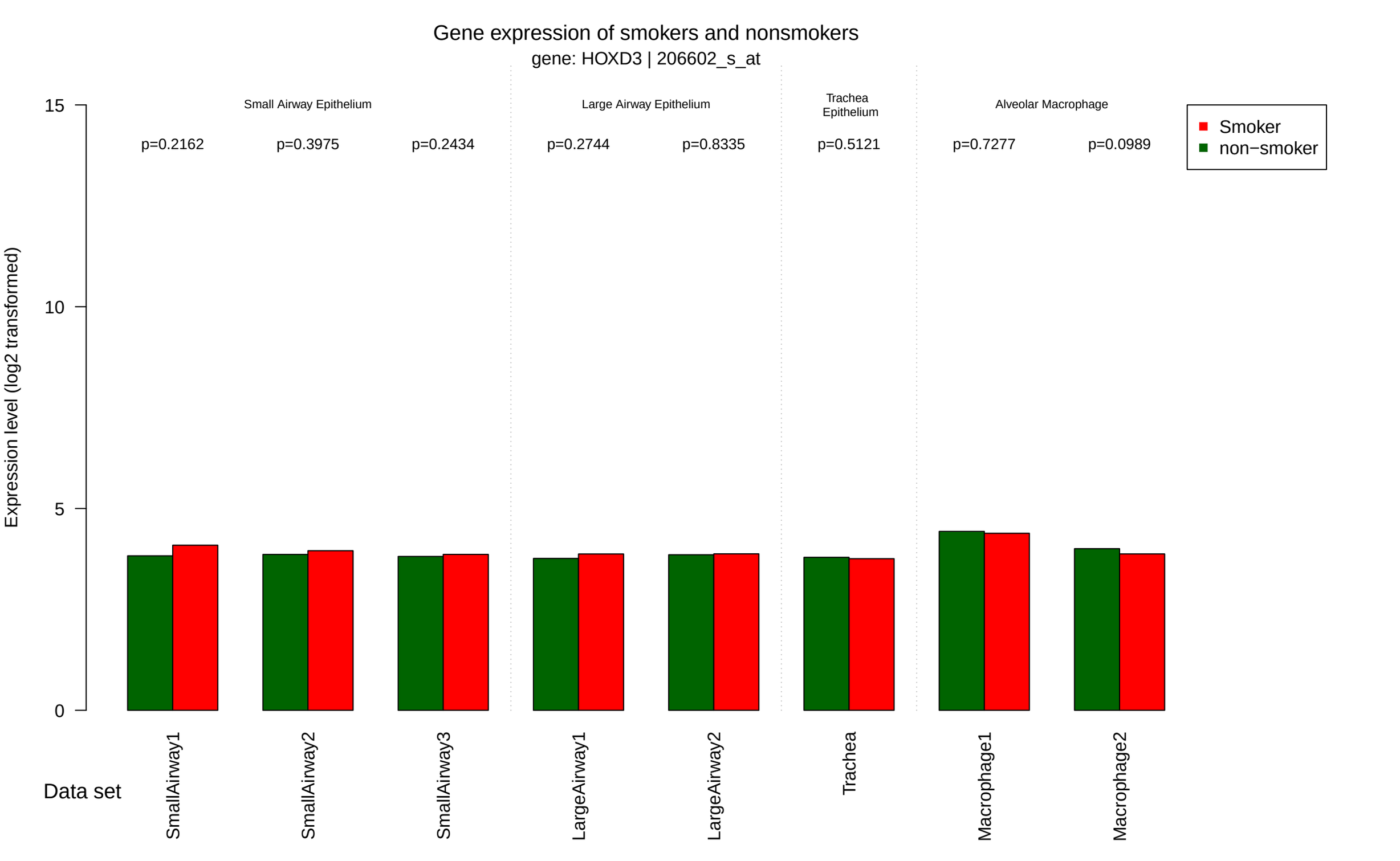
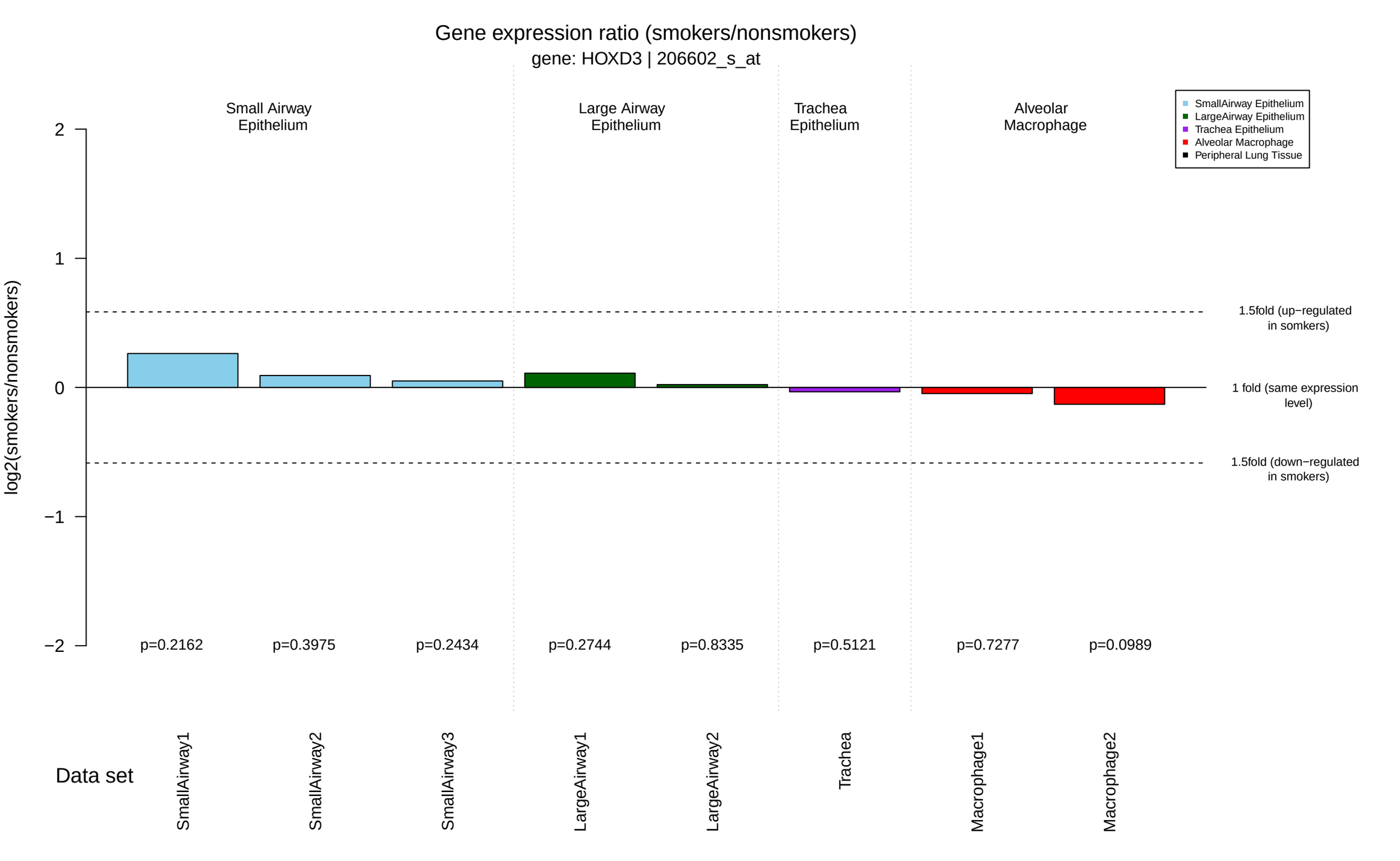
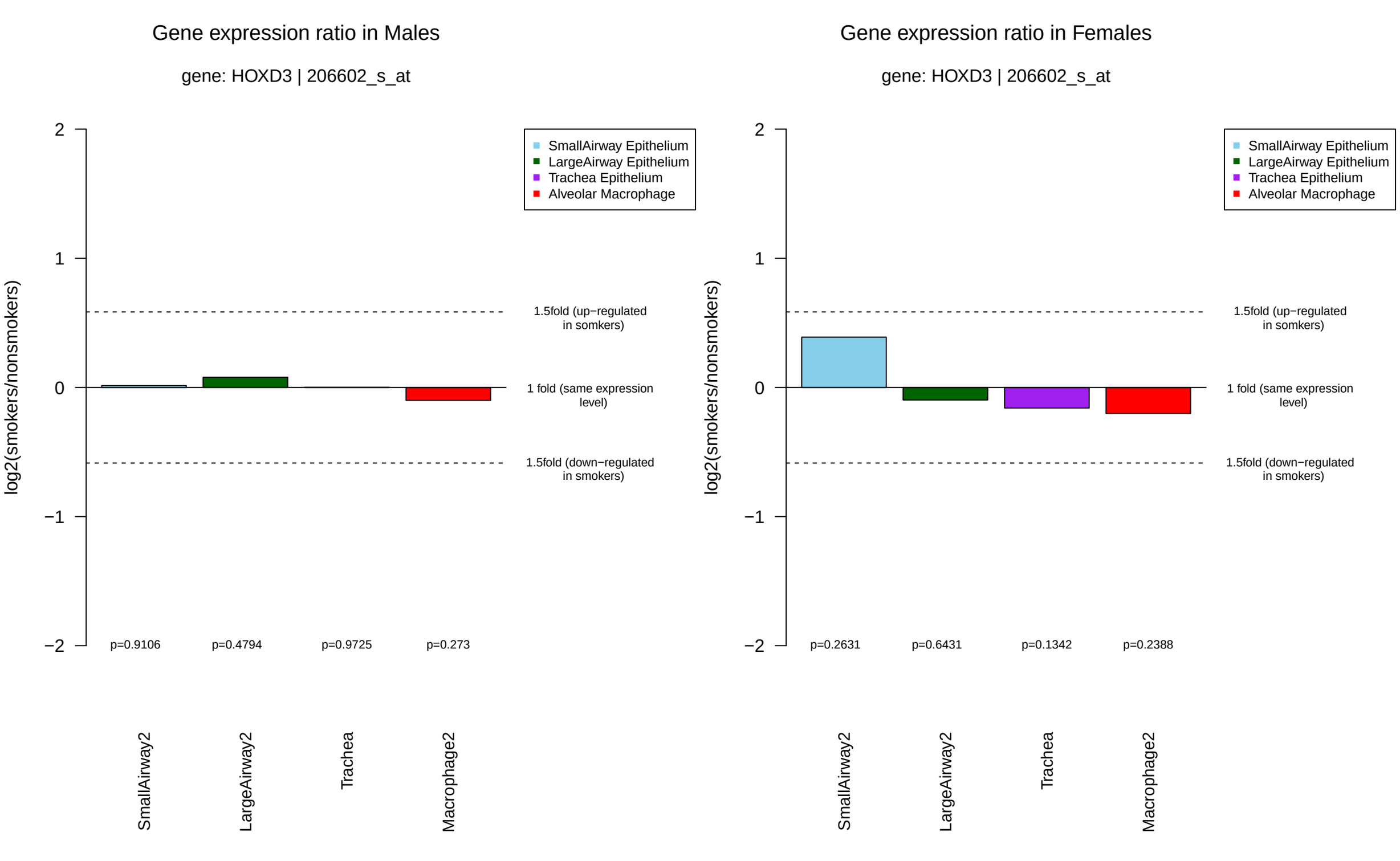
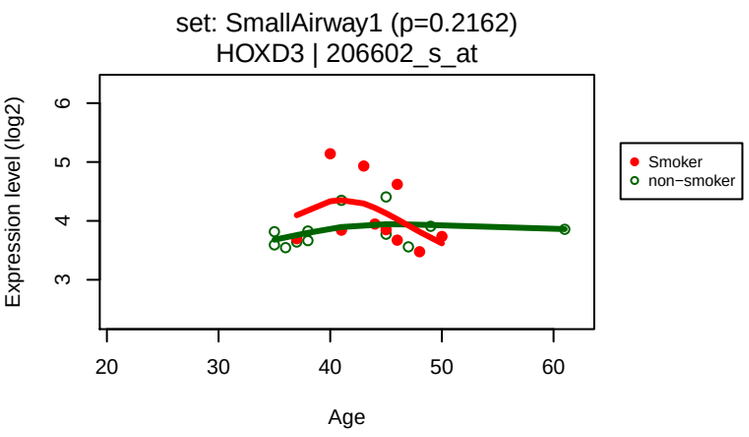
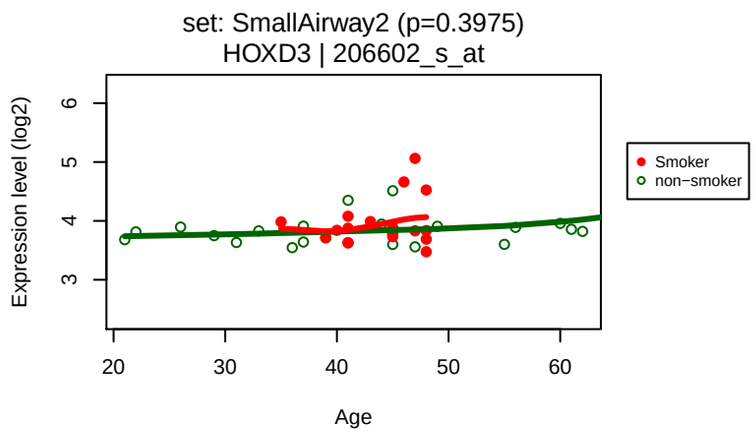
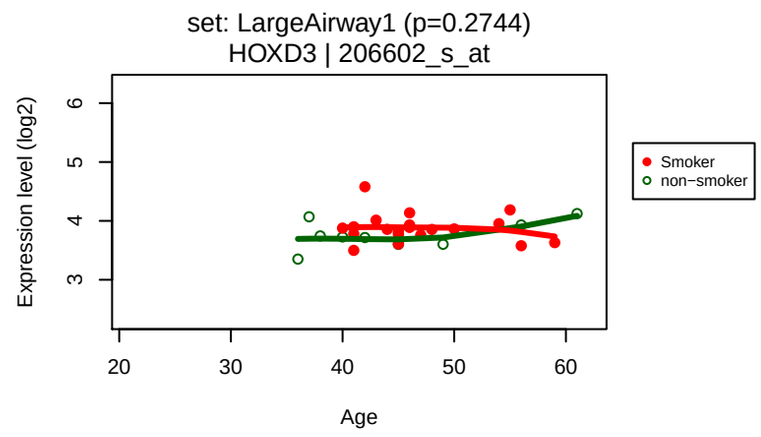

Gene expression of smokers and nonsmokers
gene: HOXD3 | 206602_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2162
p=0.3975
p=0.2434
p=0.2744
p=0.8335
p=0.5121
p=0.7277
p=0.0989
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

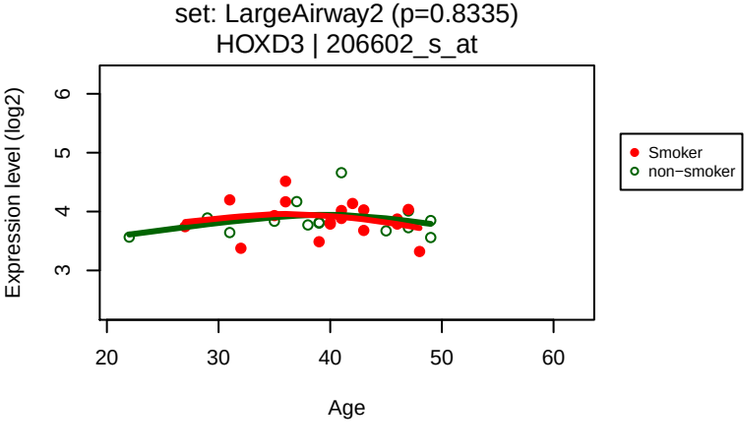

Gene expression ratio (smokers/nonsmokers)
gene: HOXD3 | 206602_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2162
p=0.3975
p=0.2434
p=0.2744
p=0.8335
p=0.5121
p=0.7277
p=0.0989
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

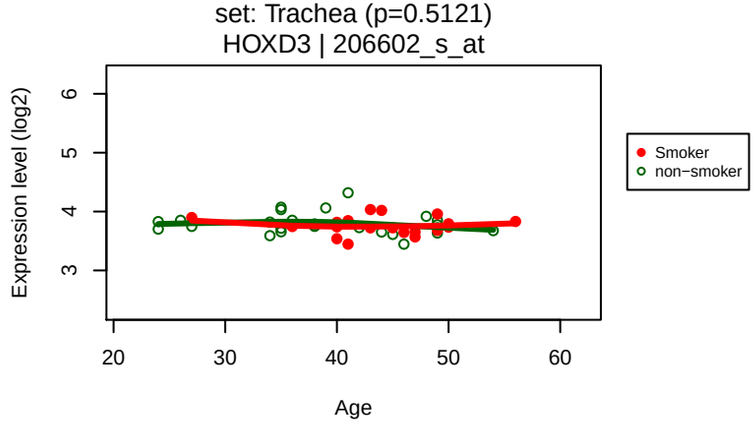

Gene expression ratio in Males
gene: HOXD3 | 206602_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9106
p=0.4794
p=0.9725
p=0.273
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXD3 | 206602_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2631
p=0.6431
p=0.1342
p=0.2388
SmallAirway2
LargeAirway2
Trachea
Macrophage2

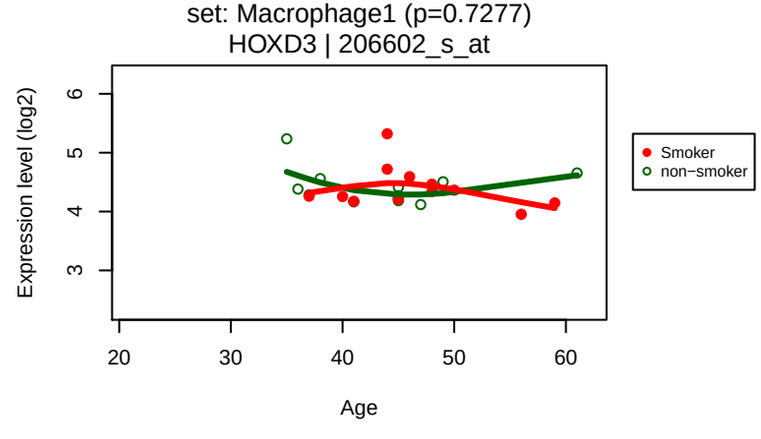

set: SmallAirway1 (p=0.2162)
HOXD3 | 206602_s_at
Expression level (log2)
Age
Smoker
non−smoker

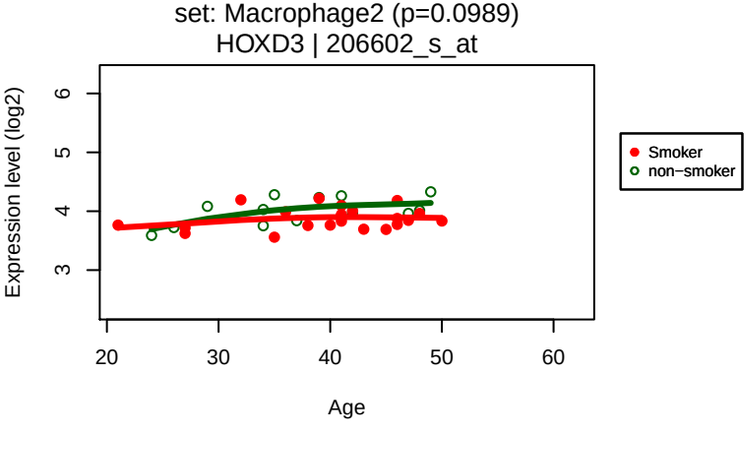

set: SmallAirway2 (p=0.3975)
HOXD3 | 206602_s_at
Expression level (log2)
Age
Smoker
non−smoker

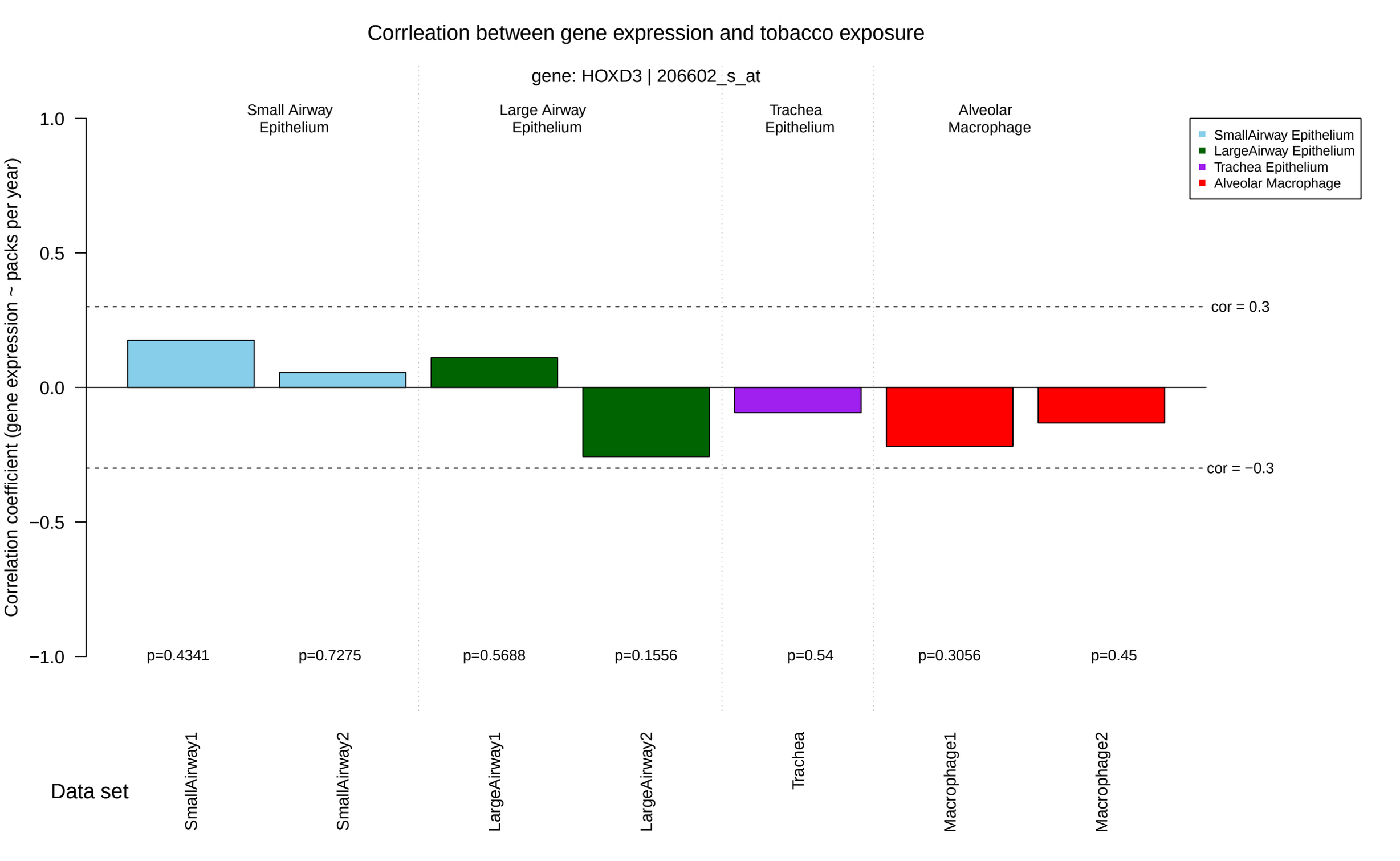

set: LargeAirway1 (p=0.2744)
HOXD3 | 206602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8335)
HOXD3 | 206602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5121)
HOXD3 | 206602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7277)
HOXD3 | 206602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0989)
HOXD3 | 206602_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXD3 | 206602_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4341
p=0.7275
p=0.5688
p=0.1556
p=0.54
p=0.3056
p=0.45
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2